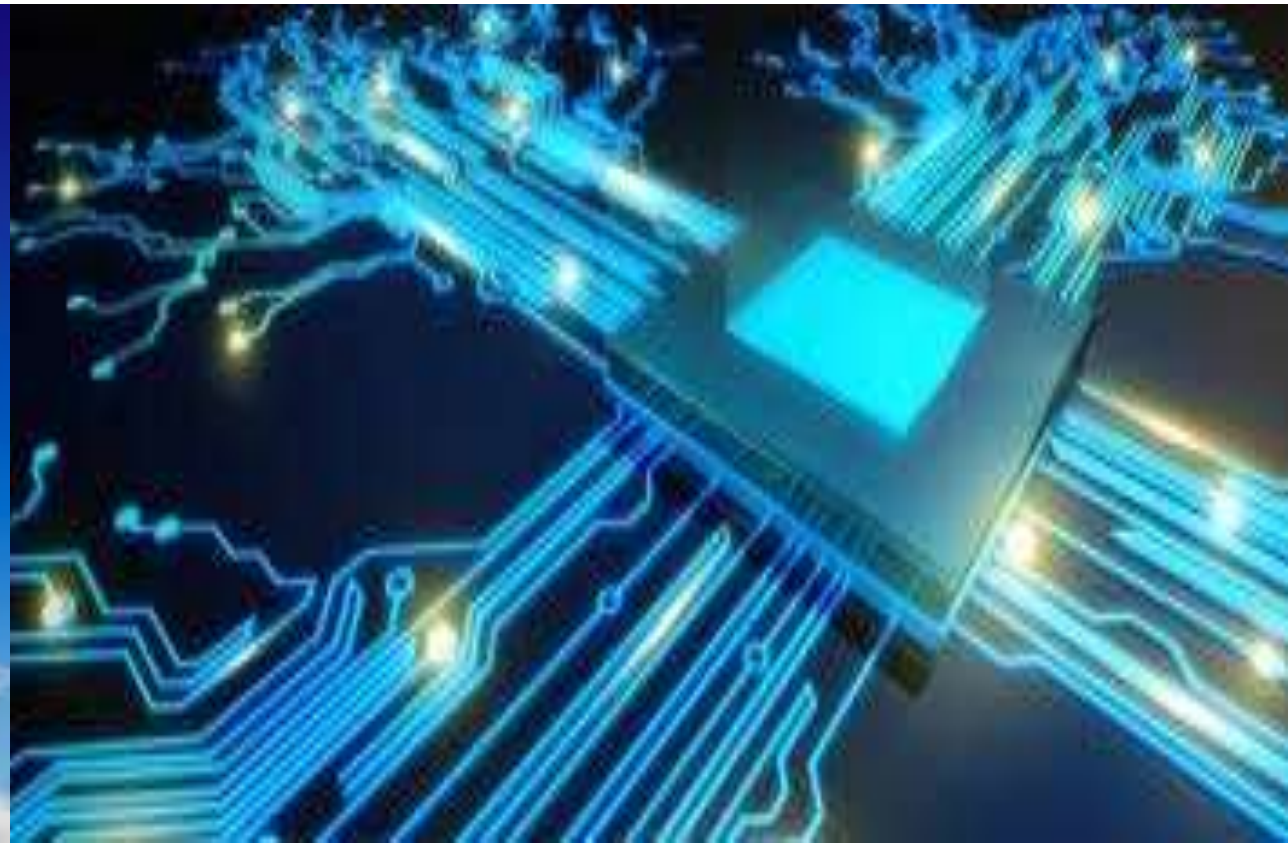
Cloud Computing





CSE 3343

PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

40 YEARS OF ACADEMIC WISDOM

MODULE – 1

Introduction to Cloud services

Contents

Course Handout Discussion

- A Facility For Flexible Computing
- From Clusters To Web Sites And Load Balancing
- Cloud Stakeholders as per NIST
- NIST cloud reference architecture
- Historical Developments
- Cloud Computing Architecture
- IaaS, PaaS, SaaS
- Types of Clouds
- Building Cloud Computing Environments
- Computing Platforms and Technologies.

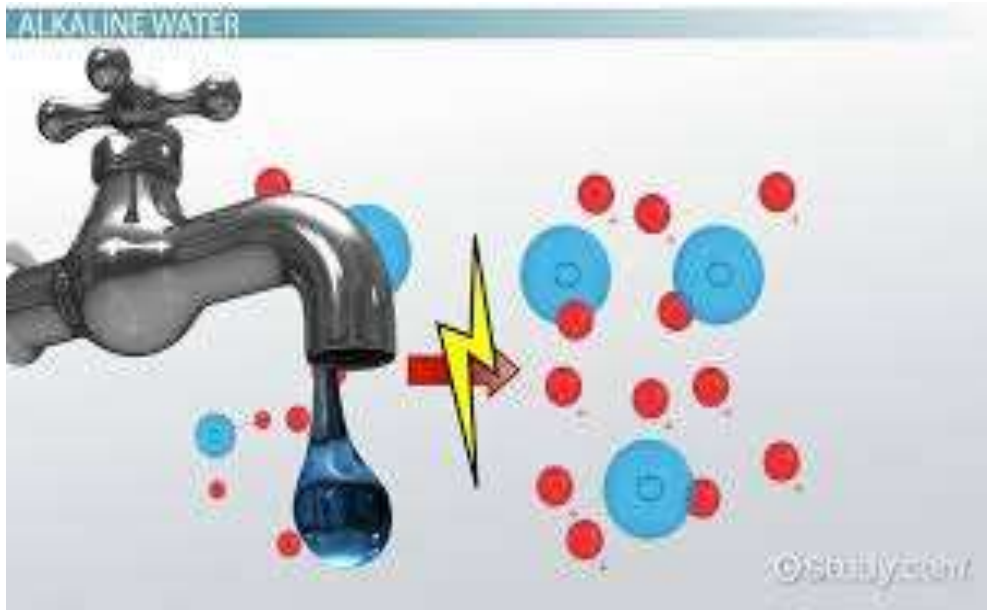
Cloud Computing



Basic services

- Water
- Electricity

1) Water



Provider



2) Electricity



Provider



Definition (Buyya – utility-oriented)

- A cloud is a type of parallel and distributed system
- consisting of a collection of interconnected and virtualized computers
- that are dynamically provisioned
- and presented as one or more unified computing resources
- based on service-level agreements
  - established through negotiation between the service provider and consumers

Cloud – one point of view

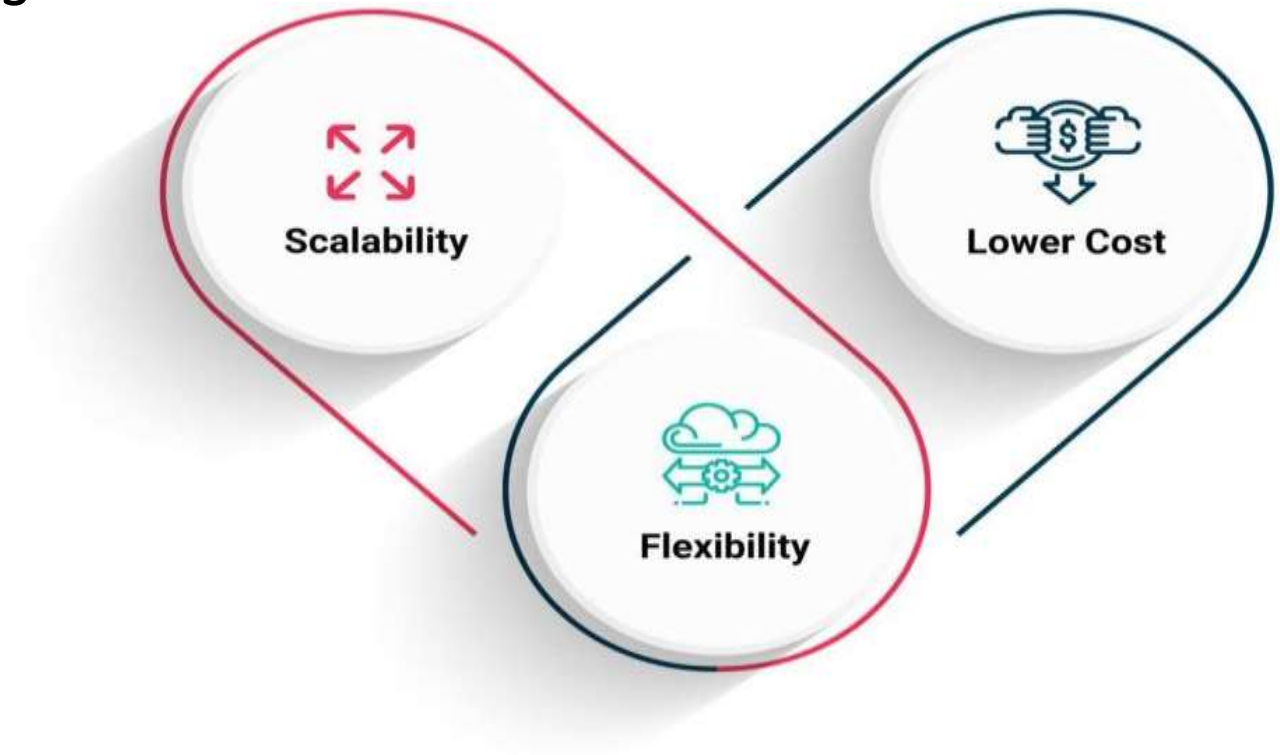
- I don’t care
- I just want them

Why is cloud popular?

1. Flexibility
2. Ensures Security
3. Maintains Productivity
4. Cost Savings and No Maintenance
5. Work From Anywhere

A facility for flexible computing

Key Aspects

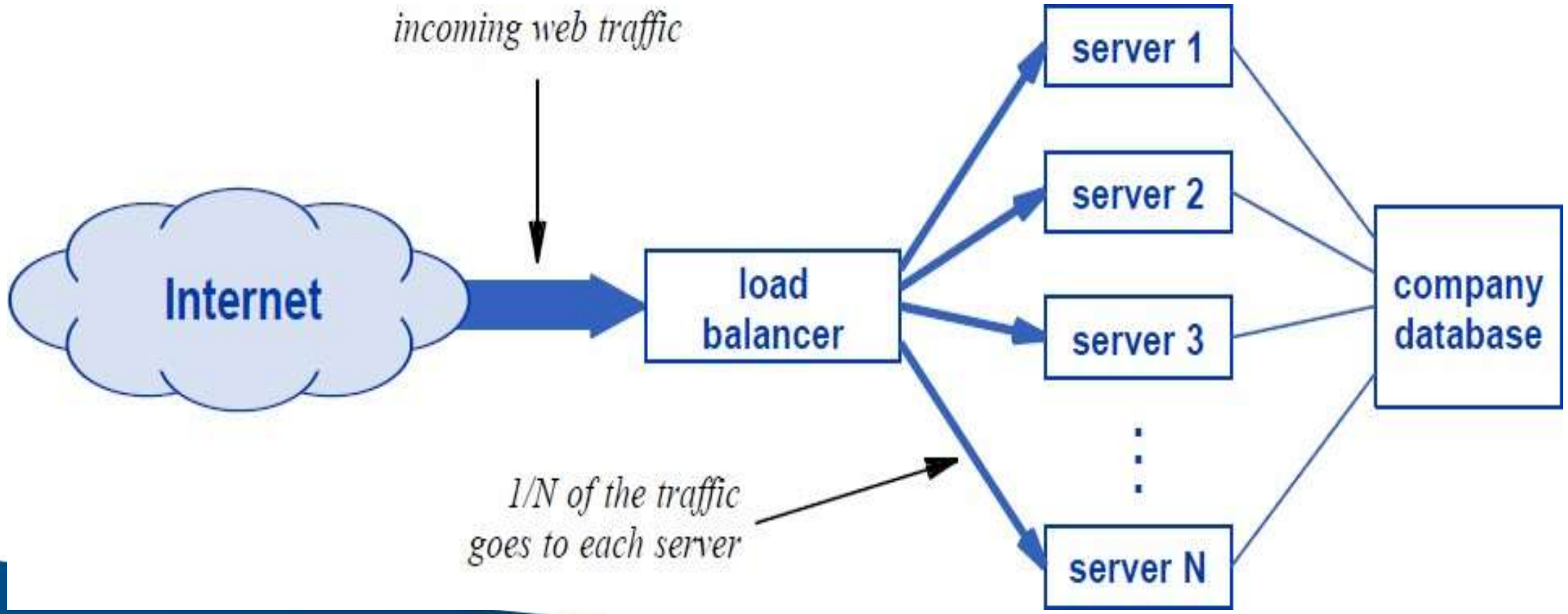


From Clusters to Distributed Web Sites

- WWW - Limitations
- Independent



Load Balancing

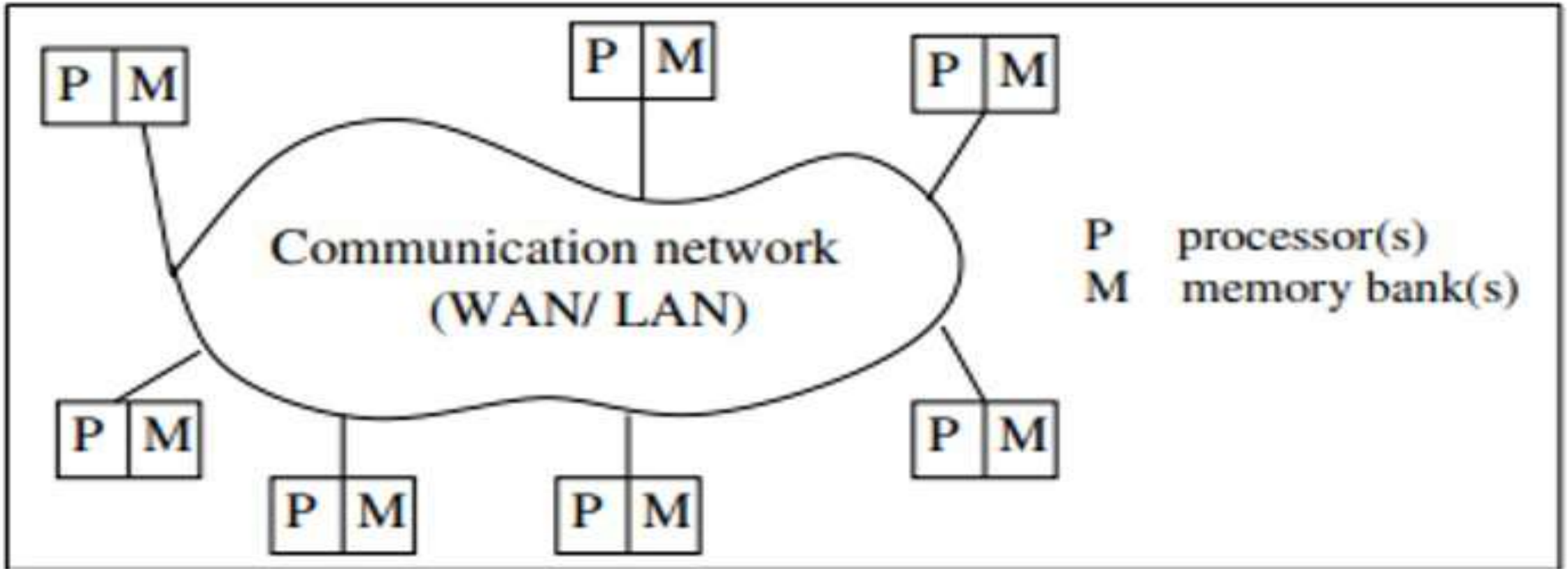


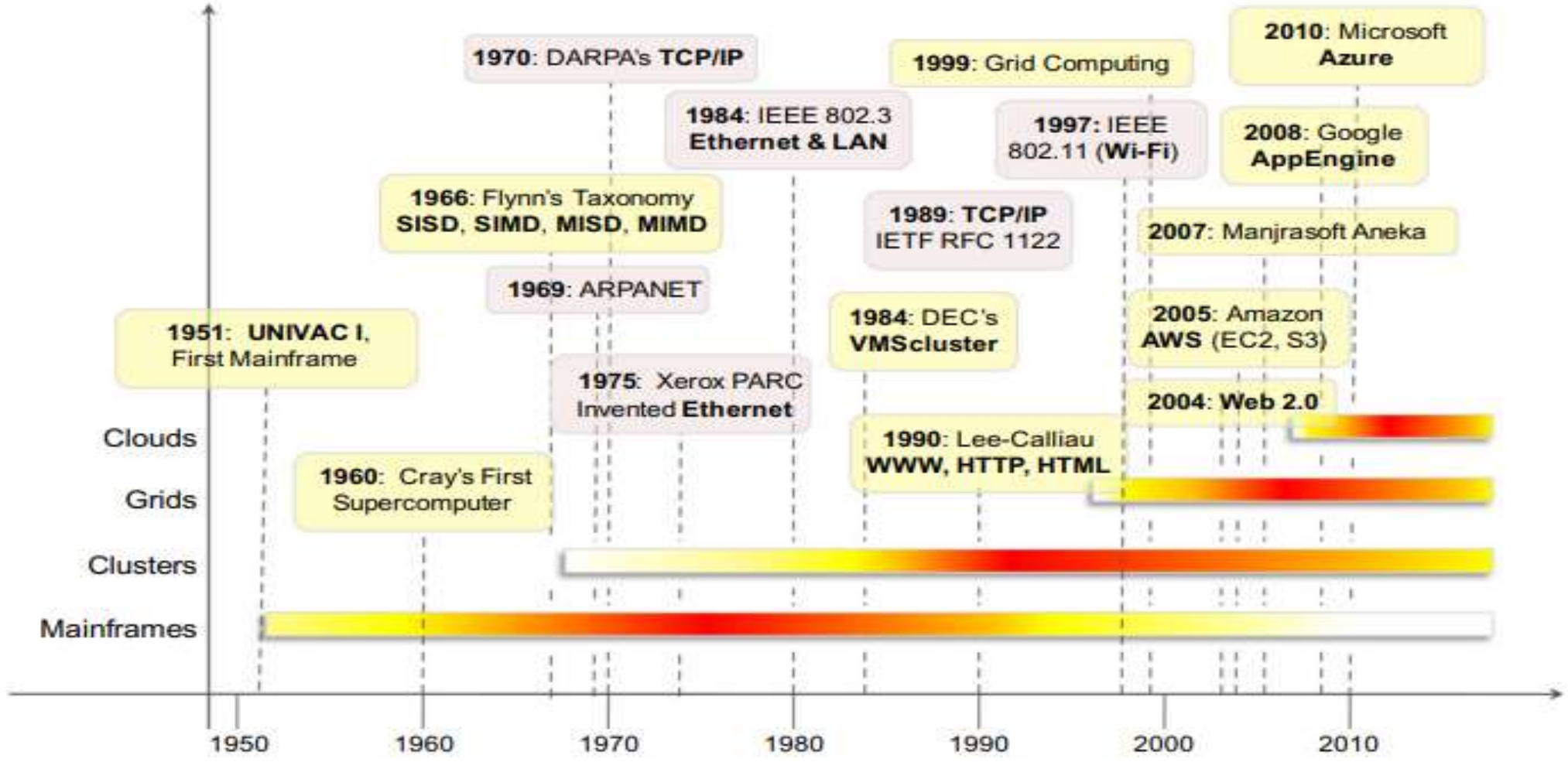
Historical Developments

5 Core Technologies

1. Distributed Systems
2. Virtualization
3. Web 2.0
4. Service-oriented Computing
5. Utility-oriented Computing

1. Distributed Systems





Evolution of Distributed Computing

Major milestones

Mainframe computing

Cluster computing

Grid computing



Mainframes

- Introduction
- Advantages
- Dis Advantages

Grid computing vs cloud computing

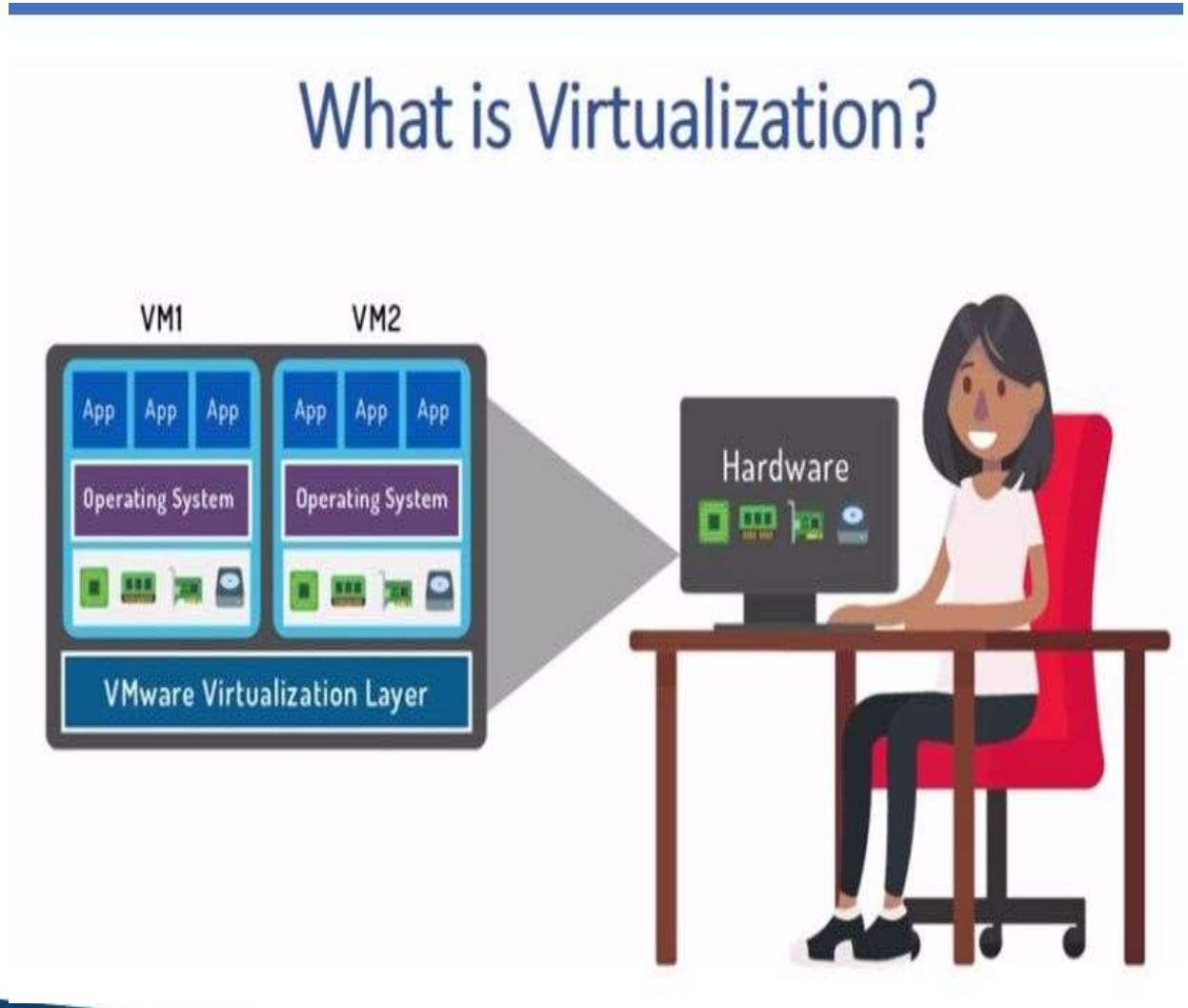
Grid Computing

- Use of widely distributed computer resources to reach a common goal
- Computing resources are distributed among different devices located in different locations
- Task is divided into several independent subtasks and each machine on the grid is assigned with a subtask
- Users can access the data in the grid computing devices via cooperate networks such as internet or a low-speed networks
- Management is decentralized
- Uses distributed architecture

Cloud Computing

- Technology that enables access to shared pools of configurable system resources and higher-level services over the internet
- Computing resources are managed centrally in data centres belonging to the cloud service providers
- Provides resources according to the requirements
- Users can access the resources through the internet
- Management is centralized
- Users' client server architecture

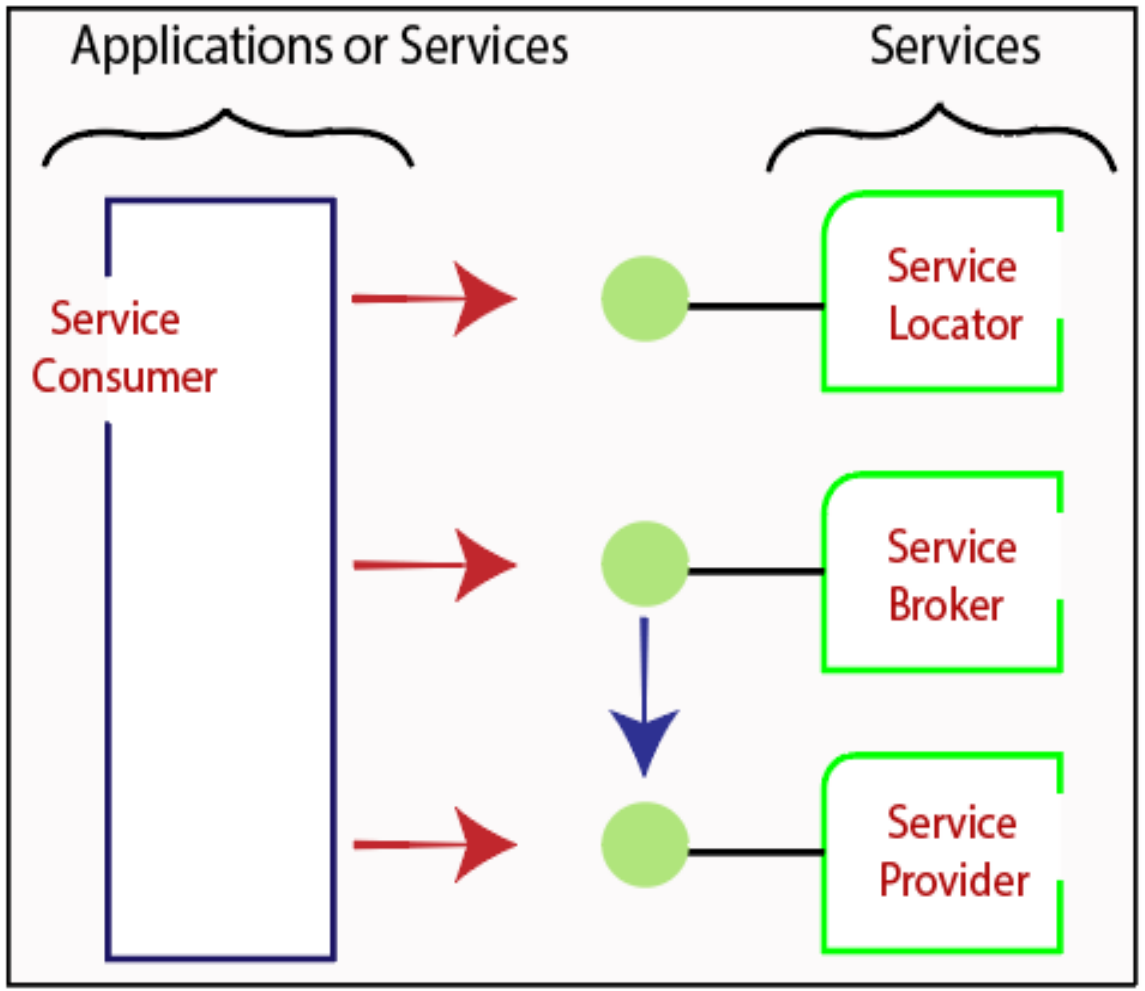
2. Virtualization



3. Web 2.0


facebook
WIKIPEDIA
hi5
YouTube
Web
2.0
LinkedIn
slideshare
twitter
flickr
Google
Apps
del.icio.us

4. Service-oriented computing



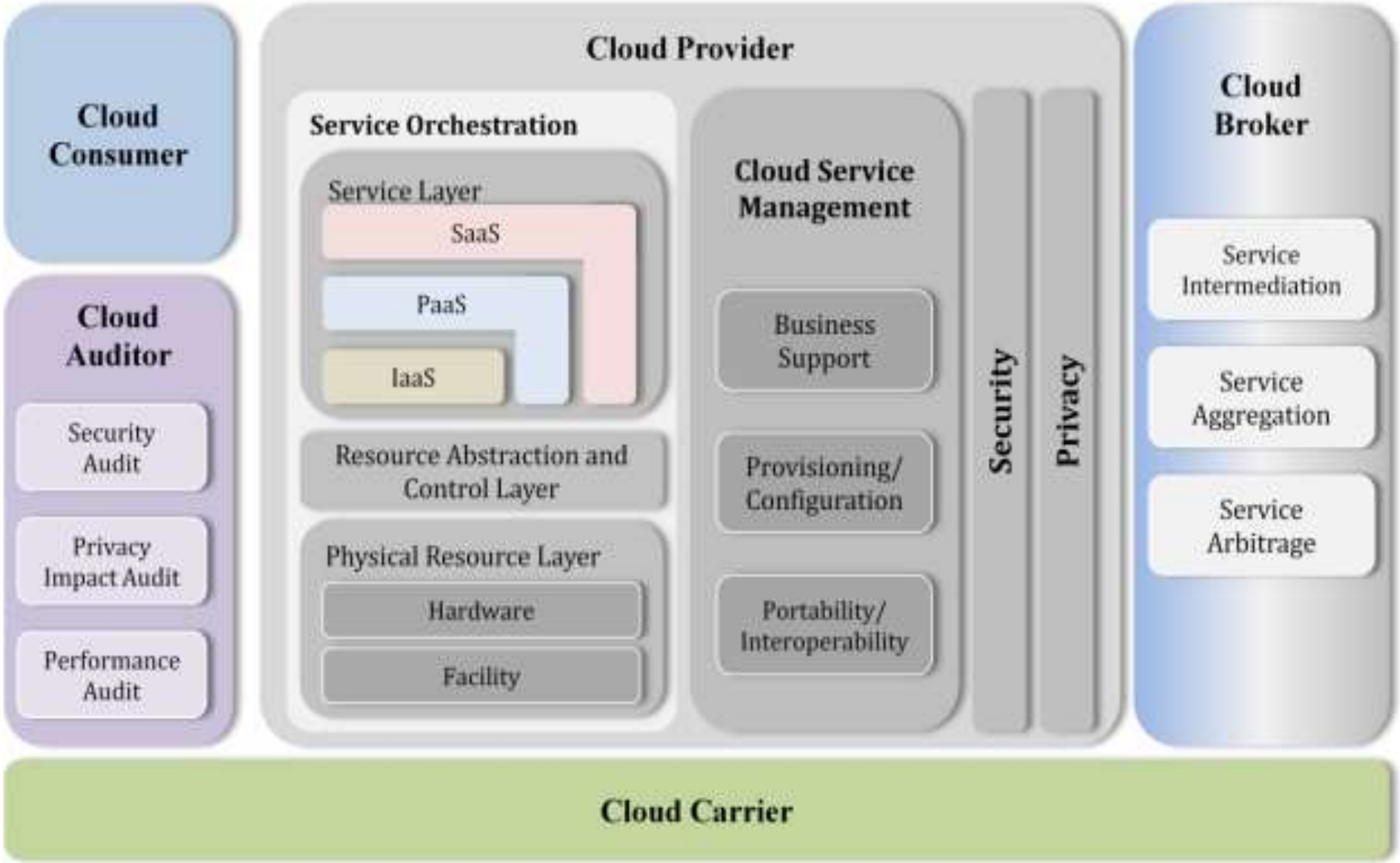
5. Utility-oriented computing

- Utility
  - Natural gas
  - Water
  - Power
  - Telephone connection

NIST cloud reference architecture

NIST - National Institute of Standards and Technology

- The NIST cloud computing - Definition
- Simple and unambiguous taxonomy of three service models available to cloud consumers:
  1. **cloud software as a service (SaaS),**
  2. **cloud platform as a service (PaaS), and**
  3. **cloud infrastructure as a service (IaaS)**


Cloud Consumer
Cloud Auditor
Security Audit
Privacy Impact Audit
Performance Audit
Cloud Provider
Service Orchestration
Service Layer
SaaS
PaaS
IaaS
Resource Abstraction and Control Layer
Physical Resource Layer
Hardware
Facility
Cloud Service Management
Business Support
Provisioning/ Configuration
Portability/ Interoperability
Security
Privacy
Cloud Broker
Service Intermediation
Service Aggregation
Service Arbitrage
Cloud Carrier

NIST cloud reference architecture

- The NIST cloud computing reference architecture defines five major actors:

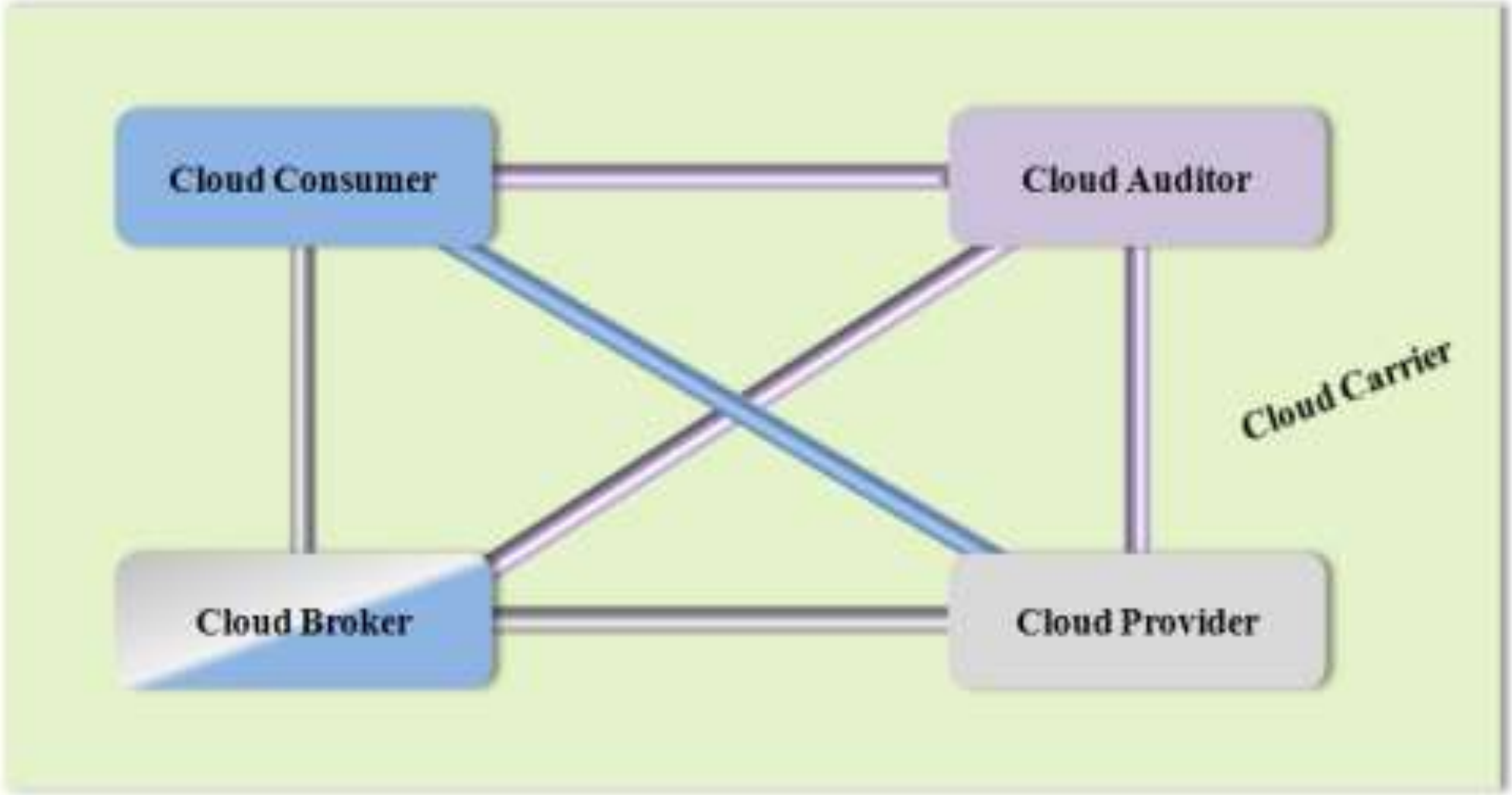
1. Cloud Consumer
2. Cloud Provider
3. Cloud Carrier
4. Cloud Auditor
5. Cloud Broker

Actor	Definition
Cloud Consumer	A person or organization that maintains a business relationship with, and uses service from, *Cloud Providers*.
Cloud Provider	A person, organization, or entity responsible for making a service available to interested parties.
Cloud Auditor	A party that can conduct independent assessment of cloud services, information system operations, performance and security of the cloud implementation.
Cloud Broker	An entity that manages the use, performance and delivery of cloud services, and negotiates relationships between *Cloud Providers* and *Cloud Consumers*.
Cloud Carrier	An intermediary that provides connectivity and transport of cloud services from *Cloud Providers* to *Cloud Consumers*.

NIST cloud reference architecture

- A cloud consumer may request cloud services from a cloud provider directly or via a cloud broker.

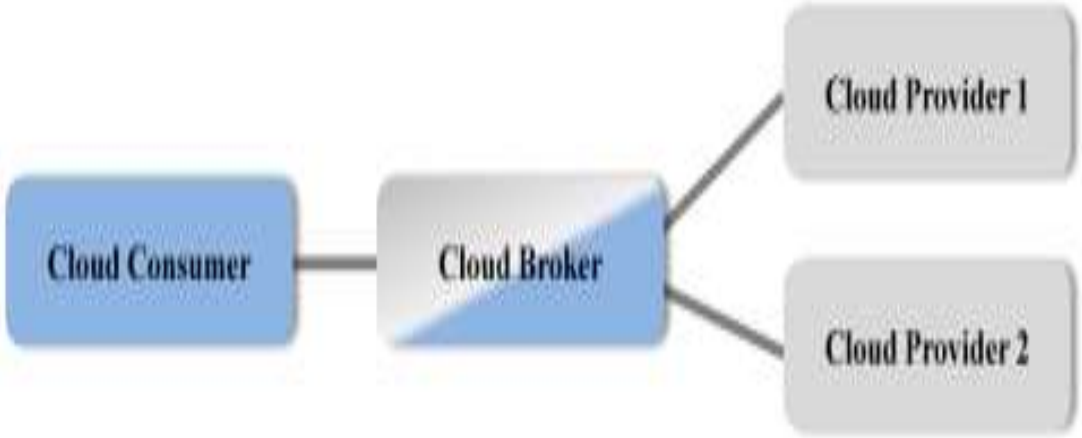
- A cloud auditor conducts independent audits and may contact the others to **collect necessary information**.


Cloud Consumer
Cloud Auditor
Cloud Carrier
Cloud Broker
Cloud Provider

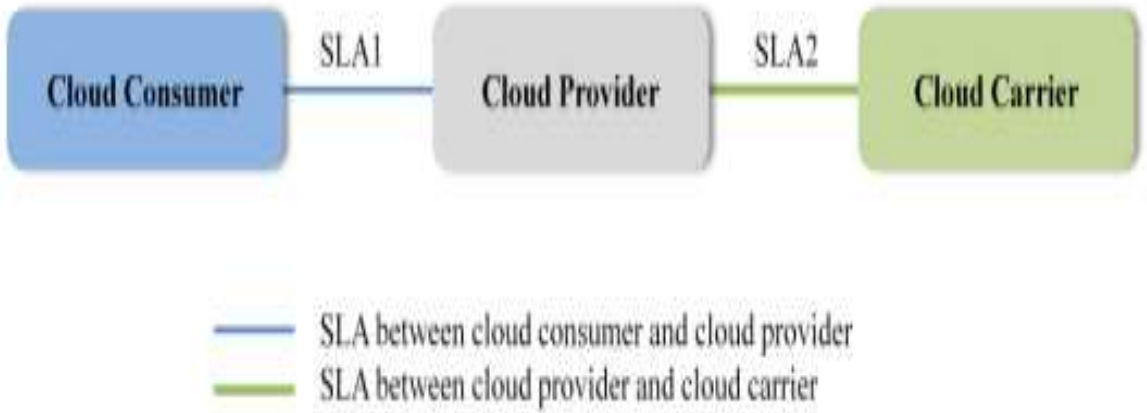
NIST cloud reference architecture with Scenario based

- Example Usage Scenario 1
- Example Usage Scenario 2
- Example Usage Scenario 3

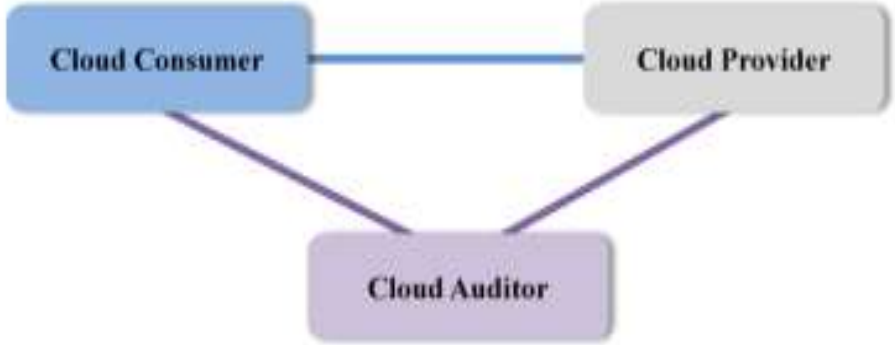
Scenario 1

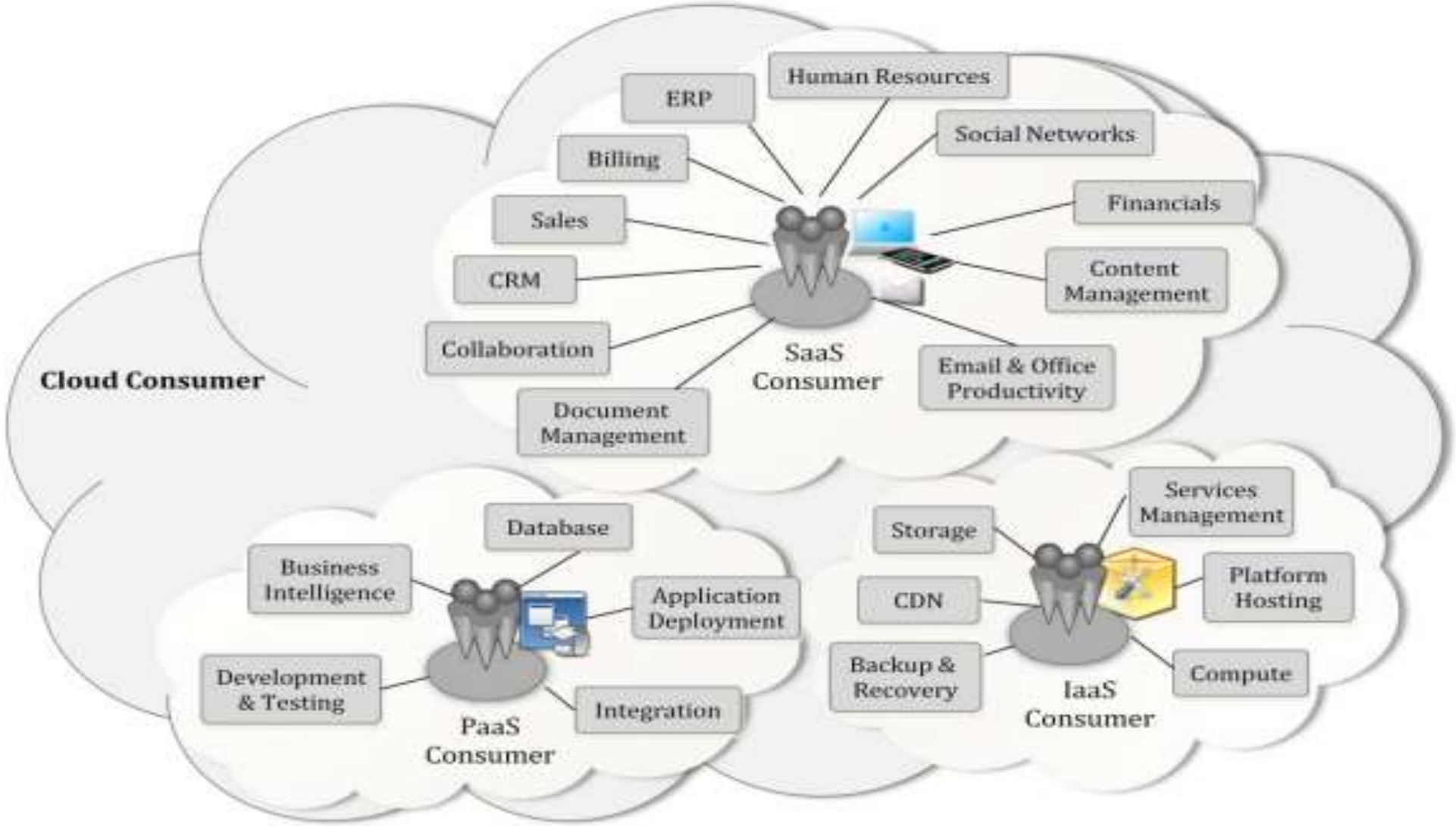


Scenario 2



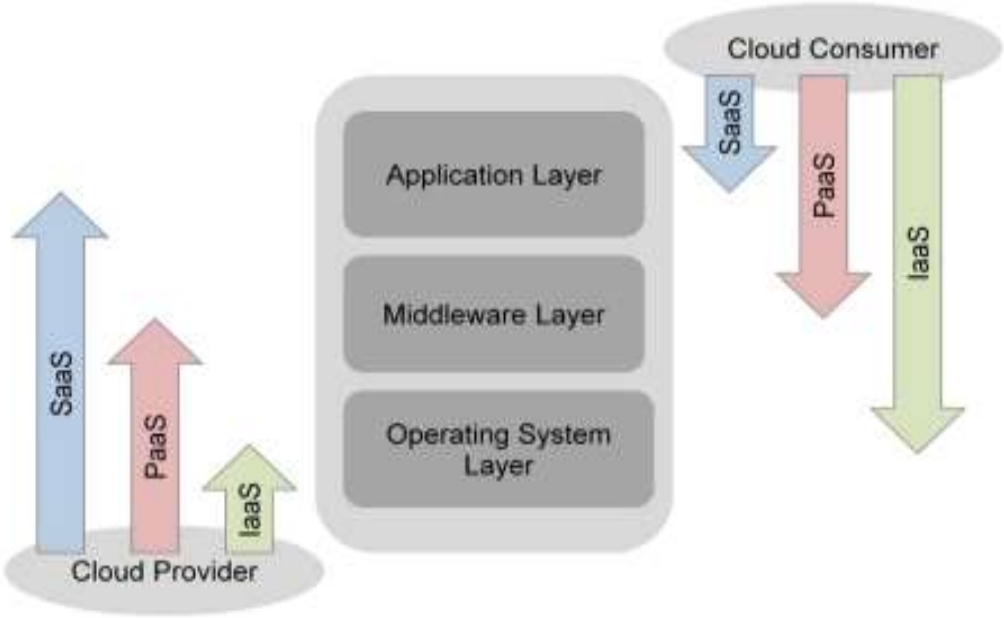
Scenario 3



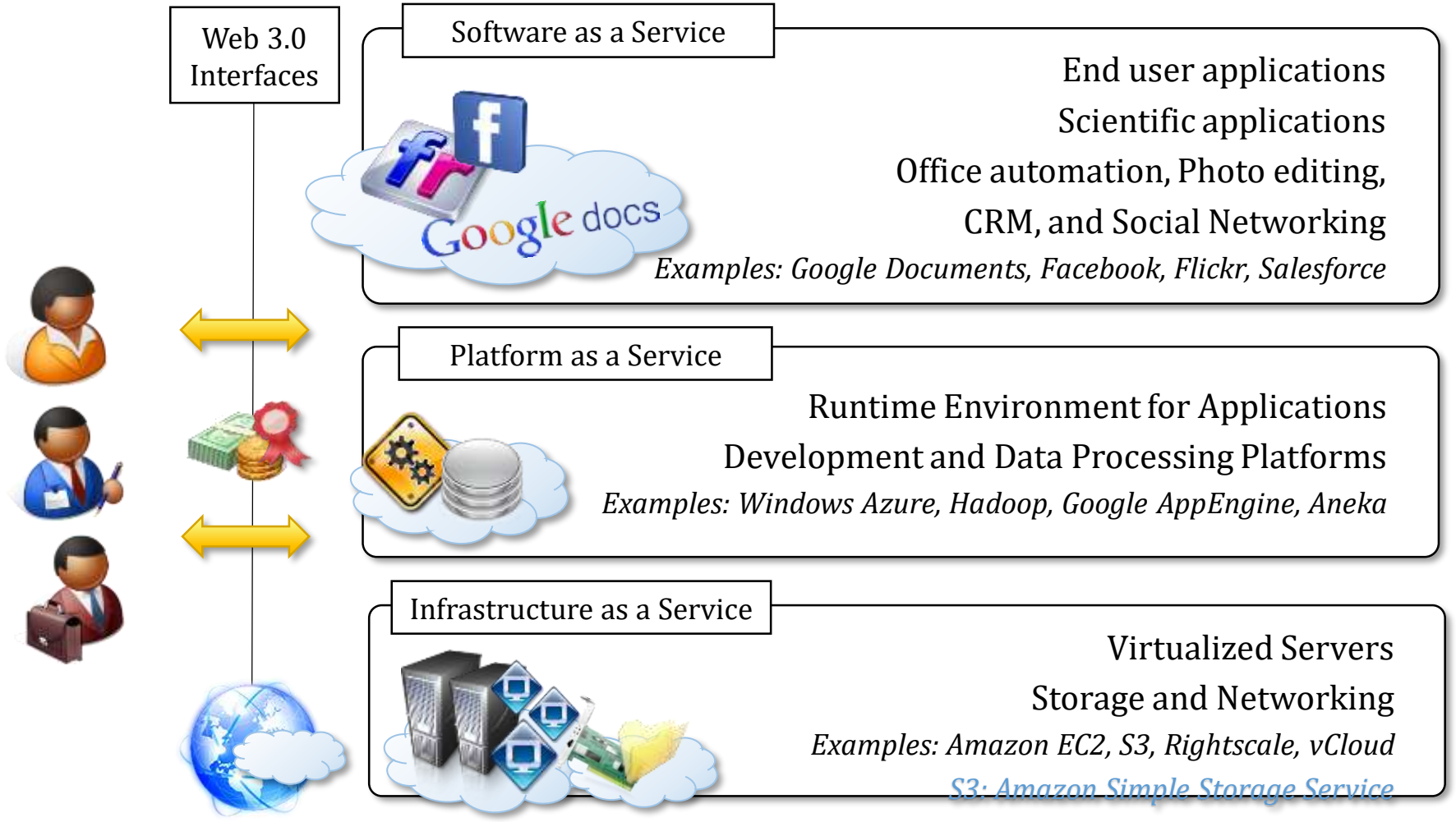


Example Services Available to a Cloud Consumer

Scope of Control between Provider and Consumer

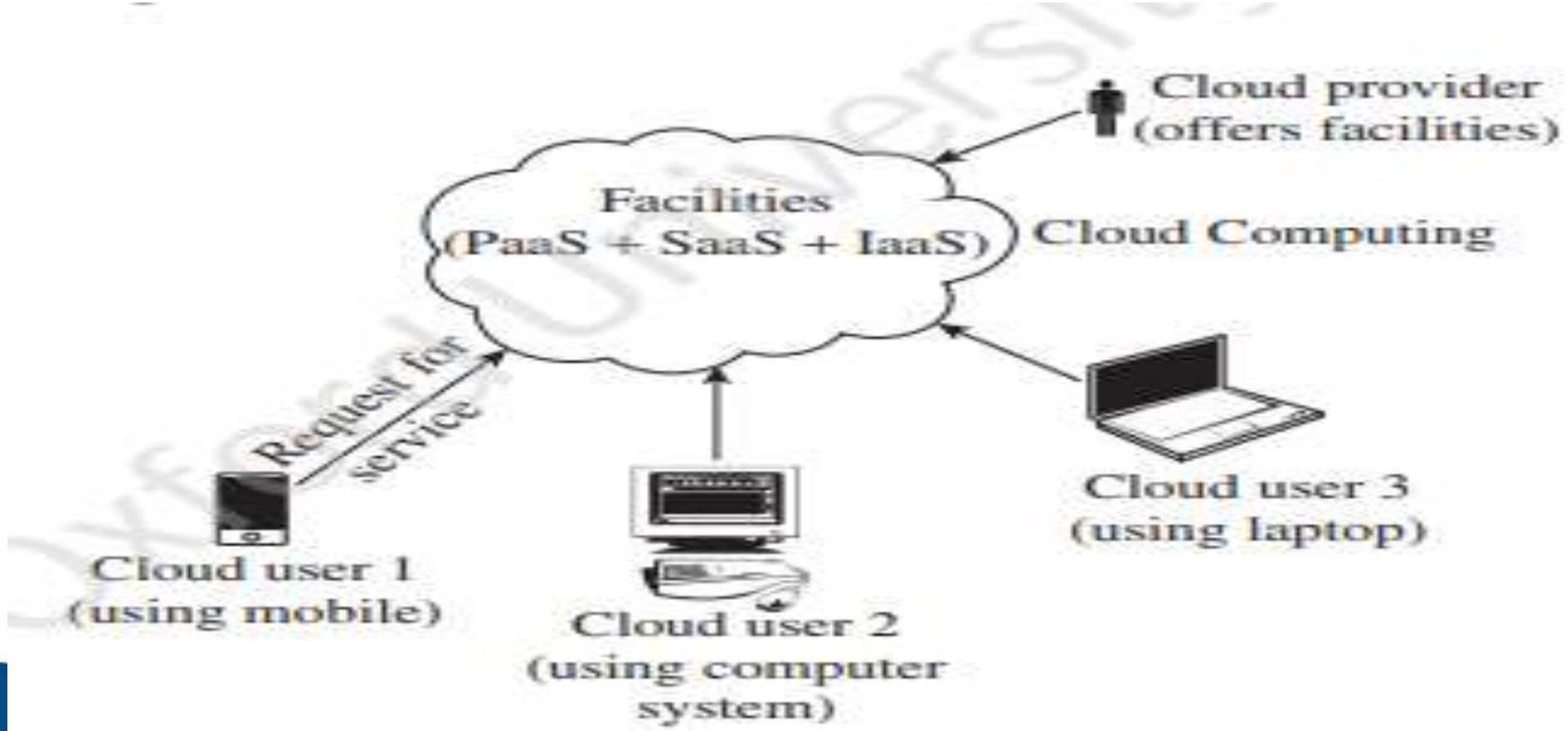


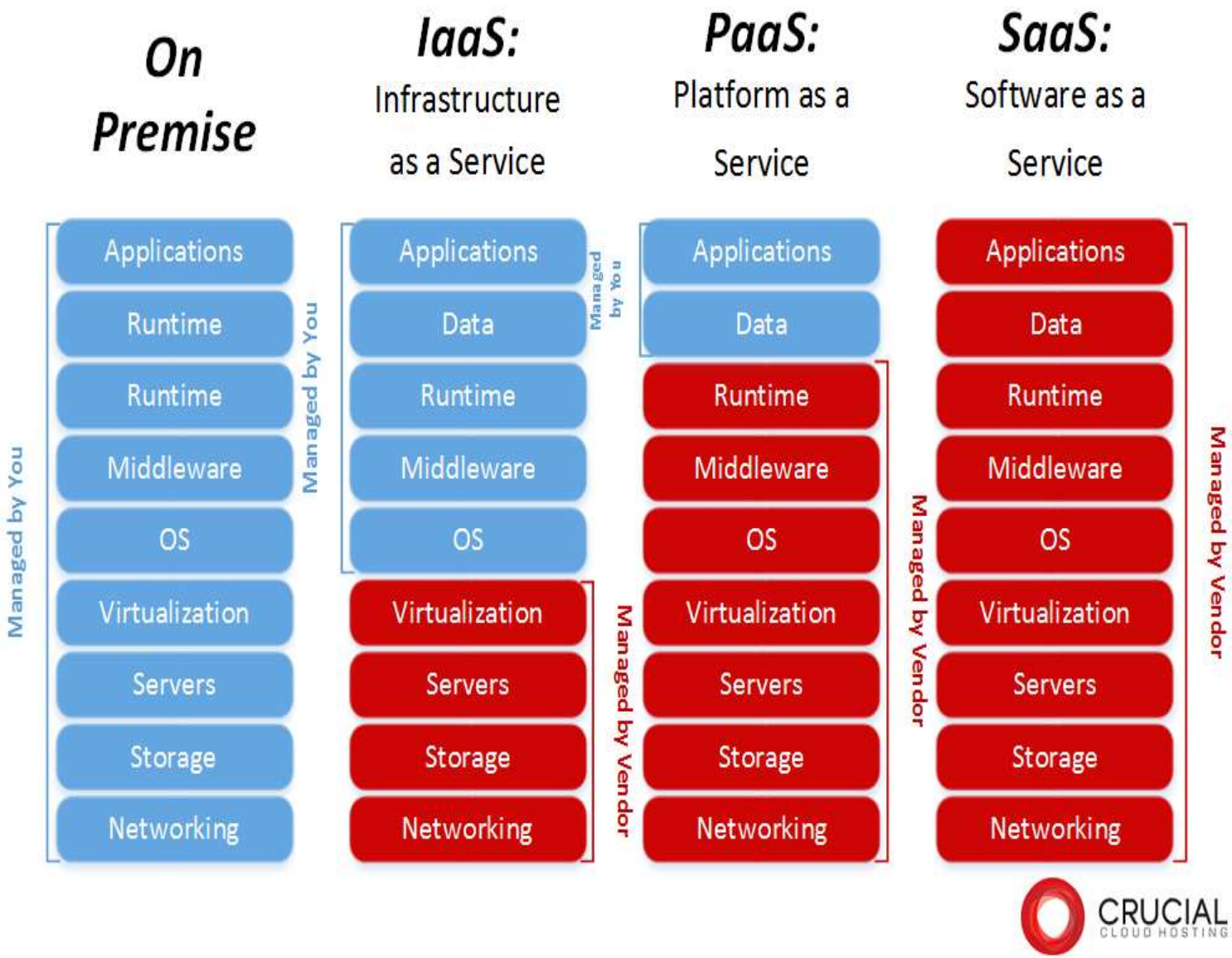
Cloud Computing SERVICE Reference Model



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40 YEARS

Basic Structure of Cloud Computing





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40 YEARS OF ACADEMIC WISDOM
CRUCIAL CLOUD HOSTING

Types of Services available in cloud computing

- IaaS --- Infrastructure as a Service
  - Characteristics
  - Advantages
  - Dis-Advantages
- PaaS---Platform as a Service
  - Characteristics
  - Advantages
  - Dis-Advantages
- SaaS – Software as a Service
  - Characteristics
  - Advantages
  - Dis-Advantages

Cloud Model	Advantages	Disadvantages	Examples
SaaS (Software as a Service)	- No need for installation or maintenance. - Accessible from anywhere with an internet connection. - Cost-effective with subscription-based pricing. - Automatic updates and security patches.	- Limited customization compared to on-premise software. - Dependency on internet connectivity. - Data security concerns as data is stored on third-party servers.	Google Workspace (Docs, Sheets, Gmail) Microsoft 365 Dropbox Salesforce
PaaS (Platform as a Service)	- Provides a development environment with tools, databases, and middleware. - Reduces time and cost for software development. - Scalable and flexible for developers. - Facilitates collaboration in software development teams.	- Limited control over the underlying infrastructure. - Vendor lock-in can be an issue. - Security risks as applications depend on third-party platforms.	Google App Engine Microsoft Azure App Services AWS Elastic Beanstalk Heroku
IaaS (Infrastructure as a Service)	- Provides virtualized computing resources (VMs, storage, networking). - Highly scalable and cost-effective for enterprises. - Full control over infrastructure and configurations. - Suitable for disaster recovery and backup solutions.	- Requires technical expertise for setup and management. - Costs can increase based on resource usage. - Security and compliance concerns as infrastructure is managed by a provider.	Amazon Web Services (AWS EC2, S3) Google Cloud Compute Engine Microsoft Azure Virtual Machines IBM Cloud

Cloud Deployment Models

- Public Cloud
  - Advantages & Disadvantages
- Private Cloud
  - Advantages & Disadvantages
- Community Cloud
  - Advantages & Disadvantages
- Hybrid cloud
  - Advantages & Disadvantages

Cloud Type	Advantages	Disadvantages	Examples
Public Cloud	- Cost-effective (pay-as-you-go) - Scalable & flexible - No maintenance required by the user - Accessible from anywhere	- Less control over security and compliance - Shared resources can lead to performance issues - Higher risk of cyber threats	AWS (Amazon Web Services), Microsoft Azure, Google Cloud Platform (GCP)
Private Cloud	- Higher security & compliance - Full control over resources - Customization as per business needs	- Expensive to maintain - Requires in-house expertise - Limited scalability compared to public cloud	VMware Private Cloud, OpenStack, Microsoft Azure Stack
Hybrid Cloud	- Best of both public & private clouds - Flexible resource allocation - Cost-efficient for dynamic workloads - Better disaster recovery options	- Complex management and integration - Security challenges in data movement - Higher operational costs compared to a fully public cloud	AWS Outposts, Google Anthos, Microsoft Azure Hybrid Solutions
Community Cloud	- Shared infrastructure for organizations with common goals - Improved security & compliance for specific industries - Cost-sharing among participants	- Limited scalability compared to public cloud - Potential conflicts in shared resource governance - May not be as cost-effective as a fully public cloud	Healthcare Community Cloud, Government Cloud (G-Cloud, AWS GovCloud), IBM Government Cloud

Major deployment models for cloud computing

Public/Internet Clouds

*Third-party, multitenant cloud infrastructure and services

*Available on a subscription basis to all



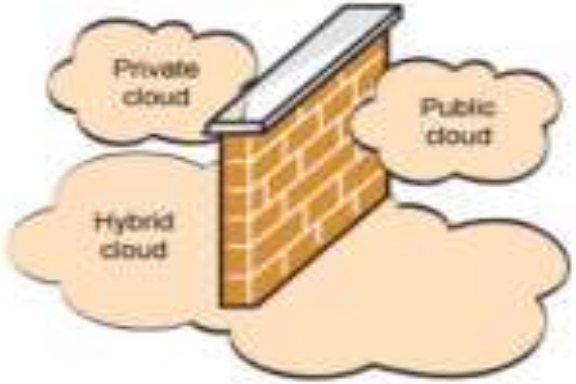
Private/Enterprise Clouds

*A public cloud model within a company's own datacenter/infrastructure for internal and/or partners' use



Hybrid/Inter Clouds

* Mixed use of private and public clouds; leasing public cloud services when private cloud capacity is insufficient



- Characteristics of Cloud
- Benefits of Cloud
- Challenges of Cloud

Aspect	Description	Example
Characteristics of Cloud Computing	Features that define cloud services	-
On-Demand Self-Service	Users can provision resources as needed without human intervention.	AWS EC2 instances can be launched anytime.
Broad Network Access	Cloud services are accessible via the internet from various devices.	Google Drive can be accessed from laptops, smartphones, or tablets.
Resource Pooling	Resources are shared among multiple customers dynamically.	A cloud provider uses virtualization to host multiple clients on the same physical server.
Rapid Elasticity	Resources can be scaled up or down based on demand.	E-commerce websites scaling up servers during peak sales seasons.
Measured Service	Cloud usage is metered, and users pay based on consumption.	Microsoft Azure bills based on virtual machine usage.
Multi-Tenancy	Multiple users share the same cloud infrastructure securely.	Dropbox stores data for multiple users securely on shared infrastructure.

Benefits of Cloud Computing	Advantages that organizations and individuals gain from the cloud	- Examples
Cost Savings	Reduces capital expenses (CAPEX) by eliminating on-premise hardware costs.	Startups using AWS instead of buying expensive servers.
Scalability & Flexibility	Businesses can expand IT resources as needed.	Netflix scales cloud resources based on streaming demand.
Business Continuity & Disaster Recovery	Data is backed up and can be restored in case of failure.	Google Cloud provides automatic failover mechanisms.
Automatic Updates & Maintenance	Cloud providers handle system updates and patches.	Microsoft 365 updates applications automatically.
Collaboration & Remote Access	Users can work together on shared files from different locations.	Google Docs allows multiple users to edit documents simultaneously.
Security & Compliance	Cloud providers offer advanced security mechanisms like encryption and IAM.	AWS Identity & Access Management (IAM) provides role-based access control.

Challenges of Cloud Computing	Difficulties organizations may face in adopting cloud	- Examples
Security & Privacy Risks	Data stored in the cloud may be vulnerable to cyber threats.	Data breaches in cloud storage services like iCloud.
Downtime & Availability Issues	Cloud services depend on internet connectivity and provider uptime.	AWS outages affecting multiple applications.
Compliance & Legal Concerns	Different regions have different data protection laws (GDPR, HIPAA).	European companies ensuring GDPR compliance in AWS.
Vendor Lock-In	Difficulty in migrating from one cloud provider to another.	A company using AWS may find it costly to shift to Google Cloud.
Latency & Performance Issues	Cloud applications may experience delays due to network dependency.	High-latency issues in cloud gaming platforms like Google Stadia.
Hidden Costs	Overuse of cloud services can lead to unexpected expenses.	Businesses unknowingly incurring high cloud storage costs due to excess data transfer.

Building Cloud Environments

Application development



What sort of applications benefit from cloud?

- Dynamically scale on demand

- Classes of applications
  1. Web applications
  2. Resource-intensive applications

Computing platforms and technologies



How to develop cloud applications?

- Amazon web services (AWS)
- Google AppEngine
- Microsoft Azure
- Hadoop
- Force.com and Salesforce.com
- Manjrasoft Aneka

1. Amazon web services (AWS)

- Compute And Storage-on-demand Services
  - Elastic Compute Cloud (EC2)
  - Simple Storage Service (S3)



2. Google AppEngine

- Google provides GAE free up to a certain amount of use for the following resources:
  - Processor Storage
  - API Calls
  - Concurrent Requests

3. Microsoft Azure

- **Three types of role:**
  - Web role
  - Worker role
  - Virtual machine role



4. Hadoop

- **Map** - transforms and synthesizes the input data provided by the user

- **Reduce** - aggregates the output obtained by the map operations

5. Force.com and Salesforce.com

6. Manjrasoft Aneka

1. **Historical Representation** – In early network diagrams, engineers used a cloud shape to depict a network (especially the Internet) where internal details were hidden, focusing only on inputs and outputs.
2. **Abstraction** – Cloud computing hides underlying complexities like data centers, servers, and networking, offering services in a simplified manner, just like a cloud in the sky hides the detailed weather systems behind it.
3. **Ubiquity & Accessibility** – Just as clouds are present everywhere in the sky, cloud computing emphasizes anytime-anywhere access to resources over the Internet.
4. **Scalability & Fluidity** – Clouds in nature change shape, expand, and contract dynamically, similar to how cloud computing scales resources up or down based on demand.
5. **Marketing & Branding** – The cloud icon is simple, relatable, and visually appealing, making it an effective symbol for businesses to convey remote and on-demand computing services.